1. Paper Details

Title:
Sign Language Recognition using Modified Deep Learning Network and Hybrid Optimization: A Hybrid Optimizer (HO)-Based Optimized CNNSa-LSTM Approach

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2. Introduction

Problem Addressed

Sign language serves as a primary means of communication for individuals who are deaf or hard of hearing. Despite its importance, effective interaction between sign language users and the hearing population remains limited due to the absence of reliable automated interpretation systems. Automatic Sign Language Recognition (SLR) is a challenging problem because it involves understanding **complex hand gestures, finger movements, and temporal motion patterns** captured in video sequences.

Traditional machine learning approaches often rely on handcrafted features, which are insufficient to capture the high variability in gesture shapes, motion speeds, lighting conditions, and signer styles. Even conventional deep learning models struggle to jointly model **spatial features** (hand shapes, orientations) and **temporal dependencies** (gesture transitions over time), resulting in reduced recognition accuracy, especially for continuous or dynamic sign language sequences.

Importance of the Research

Developing accurate and efficient sign language recognition systems has significant social and technological importance. Such systems can:

- Enable inclusive communication between deaf and hearing individuals
- Support assistive technologies in education and healthcare
- Enhance human–computer interaction through gesture-based interfaces

With the rapid advancement of deep learning, combining convolutional, sequential, and attention-based models provides an opportunity to overcome limitations of earlier approaches. This research is important because it proposes a **hybrid deep learning framework optimized using meta-heuristic algorithms**, leading to improved recognition accuracy and robustness in real-world scenarios.

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3. Methodology Overview

Proposed Model Architecture

The authors propose a hybrid deep learning framework called **CNNSa-LSTM**, which integrates multiple neural network components to capture both spatial and temporal characteristics of sign language gestures.

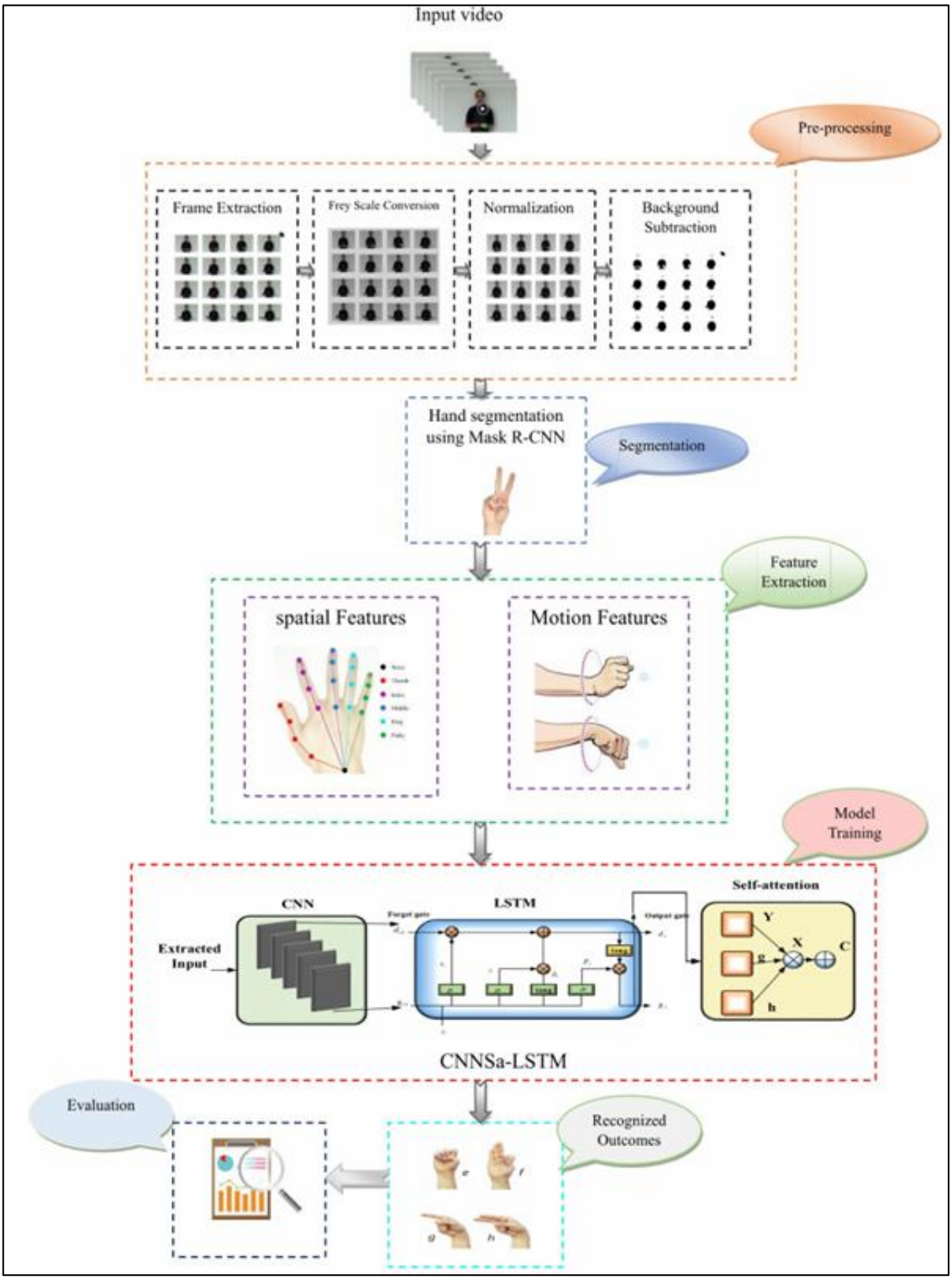

Fig: Proposed Model Architecture

1. Convolutional Neural Network (CNN)

The CNN component is responsible for extracting **high-level spatial features** from individual video frames. It captures important visual information such as:

- Hand shape
- Finger orientation
- Texture and contour patterns

A pretrained CNN architecture (such as VGG-16) is employed to leverage transfer learning, enabling effective feature extraction even with limited training data.

2. Self-Attention (SA) Mechanism

The self-attention module enhances the CNN output by assigning **higher importance to relevant spatial-temporal regions** of the gesture sequence. This allows the model to focus on discriminative features while suppressing redundant or noisy information. The attention mechanism significantly improves the model's ability to differentiate between visually similar gestures.

3. Long Short-Term Memory (LSTM)

The LSTM network models **temporal dependencies** by learning how gestures evolve over time. Since sign language consists of sequential movements, LSTM effectively captures motion continuity and temporal context, which are critical for accurate recognition.

Hybrid Optimization Strategy

To further improve performance, the authors introduce a **Hybrid Optimizer (HO)** that combines:

- **Hippopotamus Optimization Algorithm (HOA)**
- **Pathfinder Algorithm (PFA)**

This hybrid optimization approach is used to fine-tune hyperparameters such as learning rate, batch size, and network weights. Unlike traditional gradient-based optimizers, the HO strategy improves convergence stability and avoids local minima, resulting in better generalization.

Dataset and Tools

The model is evaluated using a publicly available **American Sign Language (ASL) dataset**, containing gesture images and video frames representing alphabets and digits. The dataset undergoes preprocessing steps such as:

- Frame extraction from videos
- Normalization and resizing
- Data augmentation (rotation, flipping, brightness variation)

The system is implemented using **Python-based deep learning frameworks**, such as TensorFlow or PyTorch, with additional libraries for image processing and optimization.

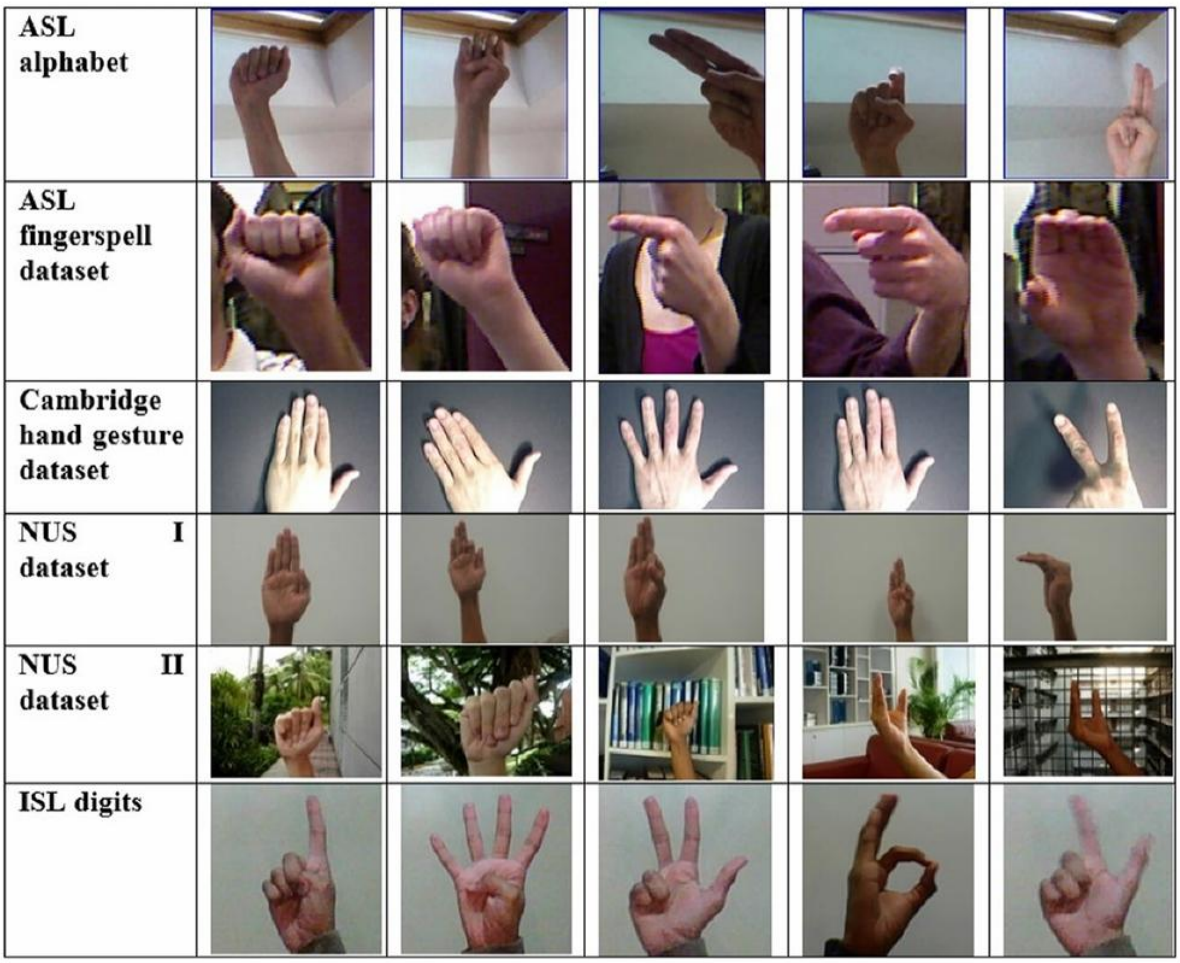

Fig. Dataset for Sign Language

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4. Key Findings

Results Achieved

The proposed CNNSa-LSTM model optimized with the Hybrid Optimizer achieves strong performance across multiple evaluation metrics:

- **Accuracy:** 98.7%
- **Precision:** 98.5%
- **Recall (Sensitivity):** 98.2%
- **F1-Score:** ~98.6%
- **Word Error Rate (WER):** 0.131
- **Sign Error Rate (SER):** 0.114

These results demonstrate a significant improvement over baseline CNN-only, LSTM-only, and non-optimized hybrid models.

Advantages of the Proposed Approach

- Efficient integration of spatial, temporal, and attention-based learning
- Reduced misclassification of visually similar gestures
- Improved robustness against gesture variability
- Enhanced training convergence through hybrid optimization
- Lower error rates compared to existing approaches

Overall, the results confirm the effectiveness of combining deep learning architectures with meta-heuristic optimization techniques.

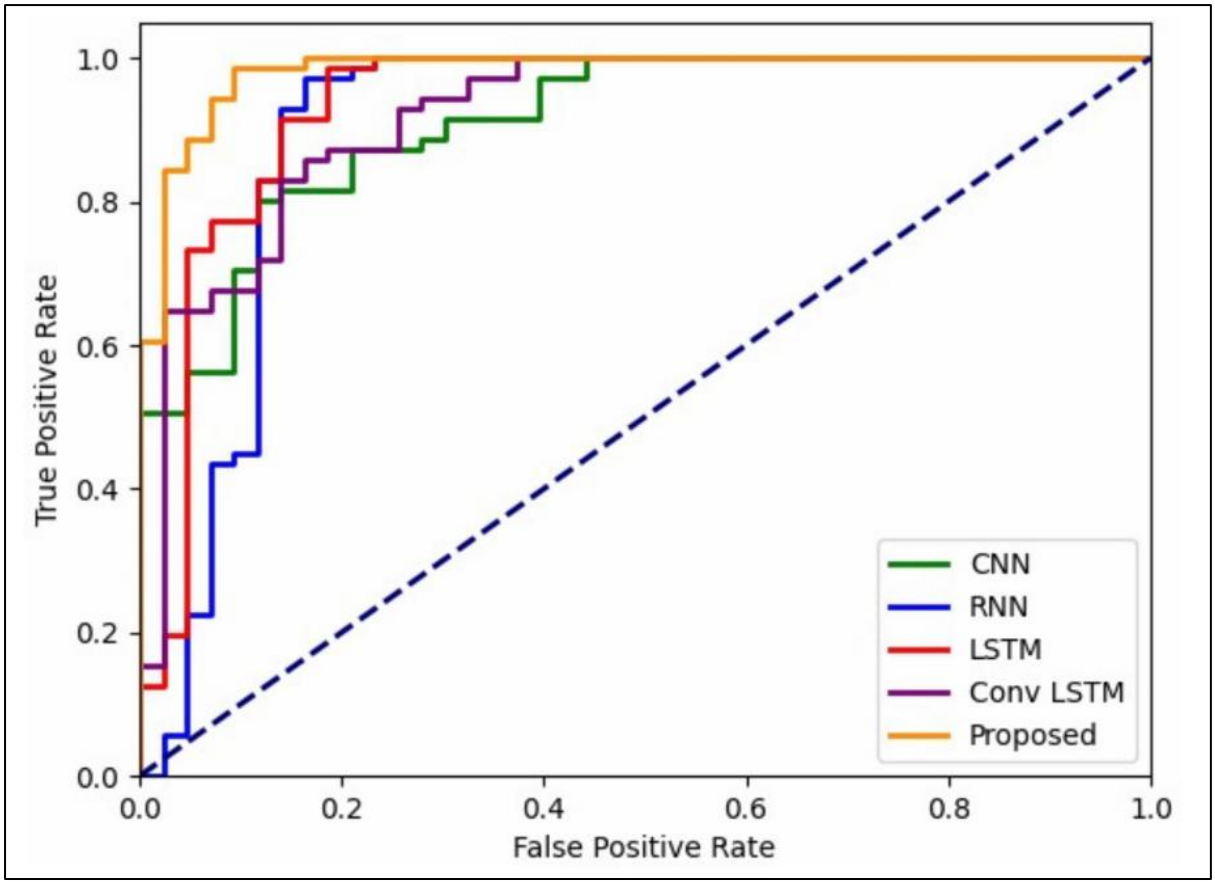

Fig. ROC curve comparison of proposed model with other models

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5. My Understanding

This paper demonstrates how complex real-world problems like sign language recognition can be effectively solved using **hybrid deep learning architectures**. The CNN extracts meaningful visual features, the self-attention mechanism highlights important regions, and the LSTM captures motion dynamics across frames. The hybrid optimizer further improves performance by tuning hyperparameters efficiently.

From this study, I learned that:

- Sign language recognition requires both spatial and temporal modeling
- Attention mechanisms significantly improve feature discrimination
- Optimization algorithms play a crucial role in maximizing deep learning performance
- Combining multiple deep learning paradigms often outperforms single-model approaches

The paper highlights the importance of architectural design and optimization in achieving high-accuracy deep learning systems.

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6. Proposed Improvements

Despite strong results, several enhancements can be explored to further advance the system:

1. Transformer-Based Temporal Modeling

Replacing the LSTM with **Temporal Transformers** could improve modeling of long-range dependencies and parallelize training, leading to better scalability.

2. Multi-Modal Feature Fusion

Integrating **hand keypoint and pose information** alongside RGB frames could enhance gesture representation and improve recognition accuracy.

3. Real-Time System Optimization

Applying model compression, pruning, and quantization techniques can enable real-time inference on edge devices such as mobile phones or embedded systems.

4. Continuous Sign Language Recognition

Extending the model to handle continuous sentence-level sign language rather than isolated gestures would significantly improve real-world applicability.

5. Robustness Enhancement

Adaptive preprocessing and noise-robust training techniques can improve performance under varying lighting conditions and complex backgrounds.

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7. Conclusion

This research presents a well-structured and effective deep learning framework for sign language recognition by integrating CNNs, self-attention, LSTMs, and hybrid optimization techniques. The proposed CNNSa-LSTM model achieves high accuracy and low error rates, demonstrating its suitability for assistive communication systems.

From a personal learning perspective, this paper provided valuable insights into hybrid deep learning design, attention mechanisms, and optimization strategies. It reinforced the

importance of combining architectural innovation with algorithmic optimization to address complex computer vision problems. The proposed improvements outline promising directions for future research, making this work a strong foundation for further advancements in accessible and inclusive AI technologies.